Discourse cues for the acquisition of the Mandarin contrafactive verb *yǐwéi*

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Abstract

In contrast to nonfactive belief verbs, such as English "think" and Mandarin *juéde* (to think/feel), the Mandarin contrafactive belief verb *yǐwéi* is negatively biased and commonly used to report false beliefs (Glass, 2023, 2025). Previous studies show that children can distinguish between nonfactives and contrafactives by 4 years of age, and the use of *yǐwéi* facilitates understanding of false-belief sentences and false-belief abilities more generally (Lee, Olson, & Torrance, 1999; Zhang & Zhou, 2022). However, it remains unclear how children learn that *yǐwéi* is contrafactive. One possible factor is the discourse context in which *yǐwéi* occurs: compared to *juéde*, *yǐwéi* is more likely to occur in contexts that highlight that the content of the attributed belief is false. In this study, we investigate whether this kind of discourse information can help Mandarin-speaking children acquire the contrafactive verb *yǐwéi* using a modified version of the Human Simulation Paradigm (Gillette, Gleitman, Gleitman, & Lederer, 1999). The results suggest that discourse might be one of the cues facilitating the acquisition of *yǐwéi*, while other linguistic cues remain to be explored.

Keywords: contrafactives; language acquisition; Human Simulation Paradigm

Introduction

Different mental state verbs express different relationships between the attitude holder, which is the subject in the matrix clause, and the embedded proposition, which is expressed in the embedded clause. Factives, such as "know p", both entail and presuppose the truth of their embedded content *p*. Thus, "know" can only be used to report true beliefs, and upon hearing utterances like (1-a), the listener can infer that the speaker takes it for granted that *it is raining outside*. In contrast, nonfactives, like "think p" in (1-b), simply assert that the agent holds the belief that *it is raining*, but it can be the case that it is not raining outside and John holds a false belief. Thus, "think" can be used to report either true beliefs or false beliefs, and *p* can be true, false, or undetermined in the Common Ground.

(1) a. John knows that it is raining outside.
b. John thinks that it is raining outside.

In English, factive verbs like "know" which are restricted to attributions of true belief (or knowledge) have no counterparts which are restricted to attributions of false belief. Such a contrafactive, like "contra p" would trigger the inference that *p* is false. On par with definitions of factives, a strong definition of contrafactivity would require that *not p* is part of the Common Ground when "contra p" is uttered, an operationalization accepted by many linguists and philosophers (e.g., Holton, 2017; Strohmaier & Wimmer, 2023, 2025). This strong definition is plausible, given the apparent lack of contrafactives in most languages. In fact, Holton (2017) argues that no Indo-European language has an attested contrafactive that is both a non-decomposable lexical unit and also presupposes the falsity of the embedded content.

Various hypotheses have been proposed to explain the apparent lexical gap for contrafactives under this stricter definition. One of them is the learnability account, which holds that contrafactives are more difficult to learn than factives. Results from a study using an artificial learning paradigm with English-speaking adults (Maldonado, Culbertson, & Uegaki, 2022) and from computational simulations (Strohmaier & Wimmer, 2023, 2025) support this learning constraint.

However, Mandarin presents a case that may fill this lexical gap for contrafactives: *yǐwéi*. For instance, in (2), we not only take the subject, Xiaoming, to believe in the embedded proposition *p*, but we also infer that *p* is false.

(2) xiǎomíng <u>*yǐwéi*</u> wàimian xiàyǔ le.
xiaoming <u>*yǐwéi*</u> outside rain ASP.
'Xiaoming <u>was under the impression</u> that it was raining outside.'

Despite its negative bias, the status of *yǐwéi* as a true contrafactive remains debated. Under a recent proposal by Glass (2025), *yǐwéi* should be treated as a contrafactive under an alternative, logically weaker definition that only requires the compatibility of the Common Ground with *not p*, and only after the assertion containing the contrafactive is accepted into the Common Ground. Therefore, *yǐwéi* enables interlocutors to consider the possibility that *p* may be false, without committing to it. Furthermore, given the speaker's choice of *yǐwéi* over other alternatives, such as *juéde* (to think/feel) or *zhīdào* (to know), any pragmatic inference that *p* is true is blocked, giving rise to the false belief inference (Glass, 2023).

In contrast to learnability studies in Indo-European languages, native Mandarin speakers have no problem using and understanding *yǐwéi*, and Mandarin-speaking children can interpret the negative bias of *yǐwéi* at the age of 4 (Zhang & Zhou, 2022). How do Mandarin-speaking children learn this contrafactive verb?

Developmental studies with Mandarin *yǐwéi*

To date, most studies involving *yǐwéi* have focused on its effect on children's false belief reasoning. For instance, Lee et al. (1999) showed that children aged 3-5 performed better on false-belief tasks when *yǐwéi* was used as part of the dependent measure. In addition, children's comprehension of *yǐwéi* correlates with their performance in false-belief tasks between the ages of 4-5 (Brandt, Li, & Chan, 2023). The existence of *yǐwéi* is commonly believed to help Mandarin-speaking children succeed in false-belief tasks, as this verb contrasts with nonfactive verbs and thus highlights that the attributed belief may be false. In a recent study investigating children's understanding of belief-reporting sentences, Zhang and Zhou (2022) found that both Mandarin-speaking adults and 4-year-olds are more likely to interpret *p* is false when embedded under *yǐwéi* than when it is embedded under *juéde*. More importantly, children comprehend false-belief attributions better when *yǐwéi* is used as the matrix verb than when *juéde* is used. Yet, it remains unclear how *yǐwéi* is learned.

Acquisition of mental verbs

Mental state verbs are challenging for children to learn. While some argue that this difficulty stems from children's conceptual constraints, others suggest that mental state verbs do not provide physically observable referents and, thus, that the mapping problem is more difficult. According to the latter view, there may be multiple sources of information needed to learn words, and children could rely on linguistic sources of information, in addition to extralinguistic ones. One common strategy that attempts to isolate the conceptual challenge and focus on the information sources during word learning is the Human Simulation Paradigm (Gillette et al., 1999). In the Human Simulation Paradigm, adults, who have mature language and conceptual systems and thus should not have conceptual constraints, are presented with a new word to simulate the learning problem faced by children. Various linguistic and extralinguistic cues can be ablated or provided to the participant to support inferences about which sources of information children might rely on during word learning.

In the case of mental verbs, even adults struggle to correctly guess mental state verbs when only extralinguistic information is present (Gillette et al., 1999; Snedeker & Gleitman, 2004; Gleitman, Cassidy, Nappa, Papafragou, & Trueswell, 2005, a.o.). Because mental state verbs are more abstract than concrete action verbs, and because verbs work as linguistic scaffolding, they benefit from their syntactic distribution during word learning, a mechanism commonly referred to as syntactic bootstrapping (Gleitman, 1990, a.o.). Moreover, while syntactic structure helps constrain the plausible interpretation of a novel word to mentalistic meanings, adding extralinguistic cues that highlight the salience of mental states, such as the presentation of a false-belief scenario, can further prompt reference to mental states (Papafragou, Cassidy, & Gleitman, 2007).

Contextual cues and discourse bootstrapping

Beyond the domain of mental state verb acquisition, studies have shown that children are sensitive to contextual information, and contextually salient events can facilitate children's learning of abstract terms. A recent study by Gomes, Huh, Yun, and Trueswell (2026) suggests that failures and violations of expectations can be "gems" to the acquisition of terms for truth-functional negation, such as "not" in English, since these cues may be rare and infrequent but transparent to the learner and extremely effective in highlighting negative meaning (Gleitman & Trueswell, 2020). This is because failures may be more valuable to mention than successful completions, and adults are also more likely to use negators to comment on them. Thus, the salience of failures may be particularly informative for learning the meaning of negators, in addition to linguistic cues.

Moreover, children are also sensitive to linguistically encoded contextual information and use discourse information during the word learning process. Sullivan and Barner (2016) provides support for the hypothesis that children use discourse information to "bootstrap" themselves into the meaning of novel words. In a similar study, children around the age of 4 can rely on discourse relations, as indicated by discourse connectives, to infer the meanings of new words, further supporting the effect of discourse on verb learning (Sullivan, Boucher, Kiefer, Williams, & Barner, 2019). Work on syntactic bootstrapping also suggests that pragmatic information is leveraged in addition to syntactic information to help the learner further constrain the hypothesis space (Hacquard & Lidz, 2019; Fisher, Jin, & Scott, 2020).

With this study, we pursue the idea that, similar to failures being "gems" to the learning of negators, discourse contexts that clearly provide information which contradicts the embedded content *p* might be one of the "gems" for the acquisition of *yǐwéi*, because false belief is usually salient in the context such that parents are more likely to comment on it and *yǐwéi* is often used to make those comments.

Current Study

Motivated by findings on the use of contextual information in word learning and the discourse bootstrapping hypothesis, we hypothesize that discourse information which contrasts with the embedded content *p* can facilitate the learning of the Mandarin contrafactive *yǐwéi p*. Although both nonfactives and contrafactives are compatible in situations where *p* is false in the discourse context, *yǐwéi* is usually preferred over the neutral alternative *juéde* .[1] We adopt a modified version of the Human Simulation Paradigm. In Experiment 1, we present participants with dialogues between adults and children where the verbs *yǐwéi* and *juéde* are removed from target

[1] *juéde* is used as the non-factive belief verb instead of the *rènwéi* because *rènwéi* is extremely infrequent in the CHILDES datasets with only 9 occurrences in child-directed speech. In addition, *juéde* can also mean "to feel" in Mandarin, but we focus on cases where *juéde* is used as a belief-reporting verb that embeds a syntactic object which expresses a proposition.

sentences. Participants are asked to recover the missing verb based on the discourse context and the rest of the target sentence. In Experiment 2, we present only the target sentence without discourse context, in order to isolate the effect of discourse information from other linguistic cues.

If the discourse status of *p* facilitates the acquisition of *yǐwéi*, Mandarin-speaking adults should be more likely to accurately recover *yǐwéi* than to incorrectly guess *juéde* when the discourse clearly supports *not p* than when there are no cues from the discourse supporting either *p* or *not p*. Additionally, when the discourse neither clearly supports *p* nor clearly contradicts *p*, participants may be more likely to choose *juéde* given its higher overall frequency compared to *yǐwéi*, even if both verbs are compatible with the dialogue.

Experiment 1: In the presence of contexts

Methods

Materials We first extracted child-directed speech that contained the two target verbs and their surrounding context from the Mandarin CHILDES corpora (Macwhinney, 2000). We limited the context presented to participants to the five lines preceding the target sentence and the two lines following it, so that each dialogue snippet contained eight lines. For subsets of the larger Mandarin *yǐwéi* and *juéde* dataset, three independent coders determined whether the discourse provides explicit information about *p*. These coders were linguistically-trained, native speakers of Mandarin.

When the discourse provides explicit information about *p*, we further coded whether it is *p* or *not p* that is supported by the discourse. For a given embedded content *p* (e.g., "It is a cat."), we considered *not p* to be supported both cases where *not p* is explicitly mentioned in the context (e.g., "It is not a cat.") and cases where the context contains some alternative proposition *q* that cannot be true at the same time as *p* (e.g., "It is a dog."). Whether a discourse provided information establishing *p*, *not p* or *q*, we used it in the "supported" condition. If the discourse provided no information about *p* nor any conflicting proposition *q*, we used it in the "unsupported" condition. Both types of contexts were considered for each verbs. In the "supported" condition, we only included dialogues where *p* is supported in the discourse context for *juéde* and *not p* is supported for *yǐwéi* (given the incompatibility of *yǐwéi p* with *p*). Thus, the combination of these two verbs and two types of discourse cues yields four critical conditions. The English translations for an example dialogue from each condition are provided in Table 1. We selected the items for this experiment from the larger annotated dataset, focusing on items where two coders agreed on the discourse feature category. 10 dialogues were identified for each condition, resulting in 40 total dialogues.[2]

An additional 10 dialogues containing either Chinese four-character idioms or adverbial temporal phrases, *yǐqián* (before/previously) and *yǐhòu* (after/in the future), were also included as controls to assess participants' language proficiency.

For critical items, we removed the target verb from the target sentence and asked participants to select between the two options, *yǐwéi* or *juéde*, in a two-alternative forced-choice. For the control items, we similarly removed the adverbial temporal phrase or two characters in the Chinese four-character idioms, and participants were instructed to select between two candidates.

Procedure Participants were instructed that a word was left blank in a dialogue between parents (or other adults) and a child, and their task was to choose between two options corresponding to what they thought the adult might have used. An example of the experimental setup is shown in Figure 1. The experiment started with four practice trials using the same task format to familiarize the participant with the task. Explicit feedback was given at the end of each practice trial. Then each participant saw all 50 critical dialogues, with order randomized across participants.

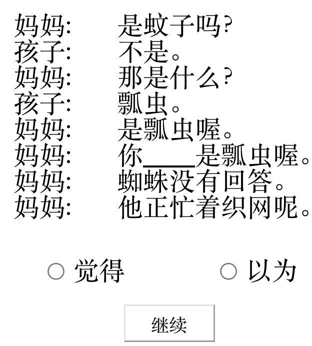

Figure 1: Example of a trial in the *juéde* + *supported* condition. The English translation can be found in Table 1.

Participants We recruited 50 Mandarin-speaking participants on Prolific. Results from 8 participants were excluded according to the preregistered exclusion criteria: 7 participants did not self-report as native Mandarin speakers, and 1 had an accuracy of 80% or less on the control items. Results from the remaining 42 participants were included in the final analysis.

Results

Figure 2(a) shows the mean accuracy in participants' guesses of the target verb. The mean accuracy across the four conditions was above chance, with *juéde* + *unsupported* and *yǐwéi* + *supported* near-ceiling, and *yǐwéi* + *unsupported* lower than in the other three conditions. In addition, there was more variability in participants' performance in the *unsupported* discourse condition than in the *supported* discourse condition, especially when the verb was *yǐwéi*.

A logistic mixed-effect regression was conducted to predict the accuracy of participants' guesses from sum-coded fixed effects of verb type, discourse type, and their interac-

[2]Data for one dialogue in the *yǐwéi* + *supported* condition were excluded in Experiment 1 due to experimenter error.

	p/not p supported in the discourse	*p/not p* unsupported in the discourse
yǐwéi	ADULT: This was really well done. ADULT: Who made this? CHILD: (This was) made by Uncle. ADULT: Oh! ADULT: It was Uncle who made it? ADULT: I *yǐwéi* it was made by Grandma. ADULT: Such a beautiful car. ADULT: Oh no!	CHILD: Right! ADULT: Then who is this? CHILD: This is the hen.. ADULT: Yeah. ADULT: The hen saw that the little sister walked over. ADULT: *yǐwéi* the little sister was going to take the chicks. ADULT: And was very angry. ADULT: What did the hen do to the little sister?
juéde	MOTHER: Is it a mosquito? CHILD: No. MOTHER: Then what is it? CHILD: A ladybug. MOTHER: Oh, it is a ladybug. MOTHER: You *juéde* it is a ladybug. MOTHER: The spider didn't respond. MOTHER: He's busy spinning the web.	MOTHER: Look, doesn't it look like an egg? MOTHER: How to put it together? CHILD: Put it together like this! MOTHER: Put it together like this. MOTHER: Right. MOTHER: I also *juéde* this puzzle is a little hard. MOTHER: Okay. MOTHER: Ah, I know.

Table 1: English translation for example dialogues in each condition. The target verbs, *yǐwéi* and *juéde*, are underlined.

tion. Also included in the model are the random by-item intercept and the random by-participant intercept and slopes for the fixed effects and their interaction.[3] The main effect of verb type was marginally significant, with the accuracy higher for *juéde* than for *yǐwéi* ($\beta = 0.53$, $SE = 0.28$, $z = 1.10$, $p = 0.0587$). However, there was no significant main effect of discourse type ($\beta = 0.11$, $SE = 0.26$, $z = 0.42$, $p = 0.6764$), and its interaction with verb type also did not reach significance ($\beta = -0.36$, $SE = 0.26$, $z = -1.40$, $p = 0.1627$). As part of the preregistered analysis, planned pairwise comparisons were conducted on the fitted logistic mixed-effect model using estimated marginal means, although the interaction was not significant.[4] When the discourse clearly supported *p* or *not p*, the accuracy did not differ significantly between the two verbs ($\beta = 0.34$, $SE = 0.75$, $z = 0.44$, $p = 0.6558$). Yet, when there was no supporting information about *p*, *juéde* had higher accuracy than *yǐwéi* ($\beta = 1.79$, $SE = 0.78$, $z = 2.30$, $p = 0.0213$).

Discussion

The relatively low accuracy of *yǐwéi* in the *unsupported* discourse condition trends in the predicted direction, suggesting that *yǐwéi* might rely more on the discourse information than *juéde*, despite limited statistical significance. The ability to detect differences between conditions might be due to a ceiling effect, as indicated by the overall high accuracy. One possible reason for this ceiling effect is that the two-alternative forced-choice experimental setup raised the likelihood of participants using the relatively infrequent verb *yǐwéi*.

In addition, 10 items per condition may be insufficient, and the analysis may have limited power at the item level. The random-effects estimates revealed substantial item-level variability ($SD = 1.37$ on the log-odds scale), substantially larger than the participant-level variability ($SD = 0.76$), suggesting considerable variability across items that might reduce sensitivity to main fixed effects. Thus, as an exploratory analysis, we fit an additional model identical to the one reported in the Results section but without the random by-item intercept. There were significant effects of verb type ($\beta = 0.30$, $SE = 0.12$, $z = 2.48$, $p = 0.0131$) and discourse type ($\beta = 0.18$, $SE = 0.09$, $z = 2.01$, $p = 0.0449$) as well as their interaction ($\beta = -0.33$, $SE = 0.08$, $z = -3.90$, $p < 0.001$). We return to this issue in General Discussion.

Setting aside the suggestive pattern, it is also possible that the discourse context is not the only information that participants used to recover the verbs. Syntactic or lexical information in the target sentence itself, such as the presence of certain adverbs or the co-occurrence of the matrix and embedded subjects, might have provided cues to the correct verb choice. Although Mandarin is morphosyntactically impoverished, past work has found that there is syntactic signal available to distinguish between different lexical classes, such as belief and desire verbs (Huang, White, Liao, Hacquard, & Lidz, 2022). Therefore, similar syntactic cues may be available to separate *yǐwéi* from *juéde*. In Experiment 2, we removed both the preceding and following context sentences to isolate the effect of discourse.

Experiment 2: In the absence of contexts

Methods

Materials and procedure In this experiment, we ablated all discourse information in order to determine how much signal remains when presented with the syntactic and lexical information in the target sentence. Only this target sentence,

[3] All regression models reported in this paper were run using the `glmer` package (Bates, Mächler, Bolker, & Walker, 2015).

[4] All pairwise comparisons of marginal means reported in this paper were estimated using `emmeans` package (Lenth, 2025).

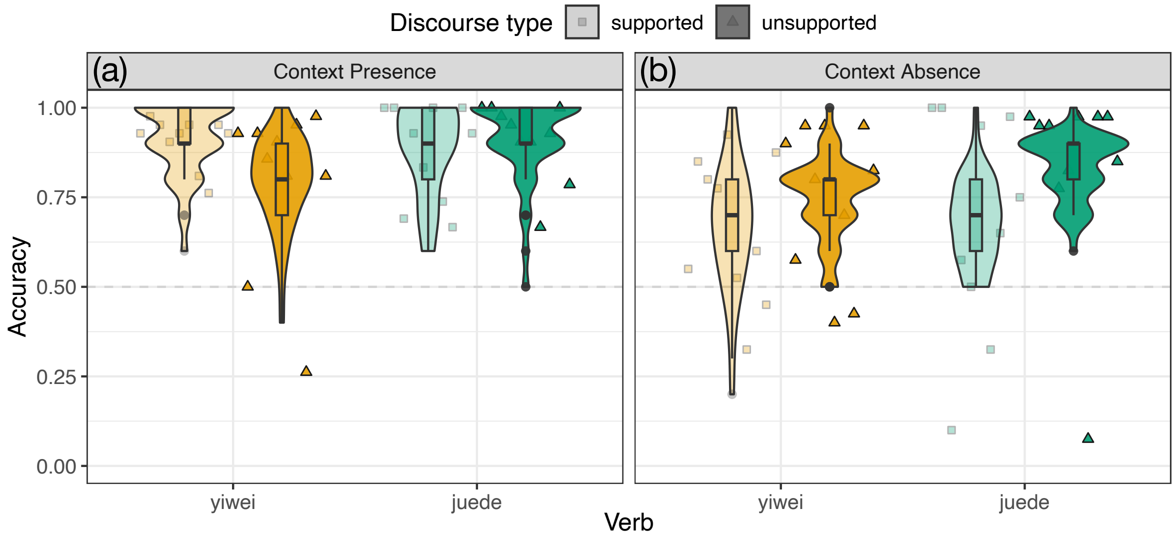

Figure 2: Mean accuracy of participants' guesses by verb and discourse types with discourse contexts (a) and without discourse contexts (b). The colored triangles and squares represent the mean accuracy for each item.

extracted from the original dialogue, was presented to a separate group of participants, allowing us to further disentangle the role of discourse from sentence-internal cues. Hence, the same set of 50 items as Experiment 1 was used, including the 40 critical items and 10 control items. The procedure was the same as Experiment 1.

Participants Fifty participants who did not participate in Experiment 1 were recruited on Prolific. Results from 10 participants were excluded based on the same preregistered exclusion criteria as Experiment 1, where 7 participants did not self-report as native Mandarin speakers and 3 participants had less than or equal to 80% accuracy on the control items. Results from the remaining 43 participants were included in the final analysis.

Results

The mean accuracy in participants' guesses of the target verb is shown in Figure 2(b). While the accuracy was above chance across all four conditions, the accuracy was higher in *juéde* + *unsupported* than that in the other three conditions. In addition, substantial variability between items was observed in all four conditions.

Using the same analysis as Experiment 1, we fit a logistic mixed-effects regression to predict the accuracy of participants' guesses from the sum-coded main effects of verb type, discourse type, and their interaction, as well as the random by-item intercept and random by-participant intercept and slopes for verb type, discourse type, and their interaction. Neither of the two main effects nor their interaction was significant (Verb type: $\beta = 0.32$, $SE = 0.27$, $z = 1.19$, $p = 0.235$; Discourse type: $\beta = -0.36$, $SE = 0.27$, $z = -1.33$, $t = 0.184$; Interaction: $\beta = -0.09$, $SE = 0.27$, $z = -0.33$, $t = 0.742$).

As in Experiment 1, this analysis may be limited by low power at the item level, given the small number of items per condition. Therefore, as an exploratory analysis comparable to the one conducted in Experiment 1, we fit a model without the by-item random intercept. There was a main effect of verb, where *juéde* had higher accuracy than *yǐwéi* ($\beta = 0.14$, $SE = 0.06$, $z = 2.24$, $p = 0.0252$). The effect of discourse type was also significant ($\beta = -0.30$, $SE = 0.06$, $z = -4.84$, $p < 0.001$), and its interaction with verb was marginally significant ($\beta = 0.12$, $SE = 0.06$, $z = -1.91$, $p = 0.0556$).

Comparison with Exp.1 As part of the preregistered analysis, we compared results between the two Experiments, with and without discourse context. Overall, participants were more likely to guess the correct verb more in the presence of the discourse context than in the absence of the discourse context. This decrease in accuracy was also more obvious in the *supported* discourse conditions than in the *unsupported* discourse conditions.

These observations were borne out statistically. We fit a logistic mixed-effect model, predicting participants' accuracy from sum-coded main effects of verb type, discourse type, context presence, and their interaction, as well as the maximal random effects structure that allowed the model to converge (random by-item intercept and slope for context presence and random by-participant intercept and slopes for verb type, discourse type, and context presence). There was a marginally significant effect of verb type ($\beta = 0.46$, $SE = 0.25$, $z = 1.84$, $p = 0.0656$), suggesting that *juéde* had higher accuracy than *yǐwéi*, which might be due to its higher overall frequency. In addition, the effect of discourse context presence was significant ($\beta = -0.61$, $SE = 0.12$, $z = -5.27$, $p < 0.001$), suggesting that participants' accuracy was higher in the presence of the context than in its absence. In addition, the interaction between discourse type and context presence was also significant ($\beta = -0.28$, $SE = 0.10$, $z = -2.81$, $p = 0.00502$), where the effect of discourse type was smaller in the absence of context than in its presence. The pairwise comparisons showed that, when the discourse supported *p* or *not p*, accuracy was much higher when participants saw the discourse context than when they didn't ($\beta = -1.79$, $SE = 0.31$, $z = -5.75$, $p < 0.0001$). The same relationship held when the discourse cue did not provide supporting information ($\beta = -0.68$, $SE = 0.30$, $z = -2.25$, $p = 0.0243$). Other main effects and interaction terms were not significant.

Discussion

Although the accuracy across all four conditions in Experiment 2 remained above chance, it was lower than in Experiment 1 with the discourse context present. In addition, the effect of the absence of the discourse context on accuracy was larger in the *supported* discourse condition than in the *unsupported* discourse condition, suggesting that discourse context is most useful in recovering the verb when it provides information about *p*.

We might have expected accuracy to drop below chance in Experiment 2 if only discourse cues mattered in distinguishing *juéde* from *yǐwéi*. However, given that adults were still more likely to recover the correct verb than not, it raises the possibility that both discourse cues and other sentence-level cues contribute to distinguishing between the verbs. We even see the suggestion of a trade-off between the two: While discourse cues are the strongest indicators of the verb choice, people might rely on the syntactic structure or lexical co-occurrence when such cues are not available, resulting in relatively high accuracy in the *unsupported* and context-absent conditions.

General discussion

In this study, we investigated how adults recover a missing belief verb in Mandarin in order to provide insight into the factors that facilitate children's acquisition of the contrafactive *yǐwéi*. Inspired by discourse bootstrapping in reference identification and the use of contextually salient failures in the acquisition of negators, we hypothesized that discourse contexts that provide information about the discourse status of the attributed belief can be effective cues, since false belief may be salient and therefore more likely to elicit comments from parents. The results show that, compared to when the discourse cue is absent or does not provide information about *p*, having discourse support for *not p* improved the accuracy of participants' guesses for *yǐwéi*, above and beyond any improvement in accuracy for *juéde*.

In addition, the results of the current study can potentially inform the learnability account of contrafactives. As mentioned in Glass (2025), it would be interesting to assess whether a weaker contrafactive verb is easier to learn than the strong contrafactives. This can be investigated using a computational simulation similar to Strohmaier and Wimmer (2025) or an artificial learning paradigm with speakers of a language that does not have a *yǐwéi*-like verb, as in Maldonado et al. (2022). Specifically, if Mandarin-speaking children rely on discourse cues to learn *yǐwéi*, will models and adult speakers of other languages be more likely to entertain weak or strong contrafactives when trained on a novel verb with *not p* cues in the discourse?

Despite these suggestive results and implications, there are a few limitations of the current study. As evident in both experiments, accuracy was high across the board. One possible reason for this ceiling effect is the choice of the dependent measure. Since *yǐwéi* is far less frequent than *juéde*, having *yǐwéi* as an option in a two-alternative forced-choice experimental setup might increase its usage by participants. Future studies can use an open-ended dependent measure, requiring participants to fill out the missing word, similar to the original Human Simulation Paradigm design. Participants might then generate a wider range of possible responses, including other nonfactive verbs like *rènwéi* (to think) and factive verbs like *zhīdào* (to know). It would be informative to examine whether *yǐwéi* remains relatively preferred when discourse supports *not p* once the potential advantage from the forced-choice design is removed.

As argued in prior literature, multiple sources of information can serve as cues for hypothesizing the meanings of new attitude verbs (Gleitman et al., 2005; Papafragou et al., 2007), and discourse information constitutes only one of the many sources available to children. In the case of *yǐwéi*, these results suggest that the discourse context is useful to the learner, but that other cues may also facilitate learning. As a preliminary step towards identifying potential sentence-level cues, we analyzed the 40 target sentences used in this study. 12 of the 20 sentences with *yǐwéi* had a first-person pronoun as the matrix subject, and all reported false beliefs about others. In contrast, *juéde* had 8 sentences where a first-person pronoun was used as the matrix subject, and 2/8 were cases where the speaker reported a belief about themself. In addition, *yǐwéi* occurred only once with a second-person subject in the discourse *unsupported* context, much less frequently than *juéde*. While this is limited to the 40 sentences used in this study, an ongoing corpus study aims to analyze a wider range of information, including the pairing between certain modifiers and *yǐwéi* (e.g., *běn yǐwéi*, originally thought) and the use of different matrix and embedded subjects with *yǐwéi* for different discourse functions. Armed with more information about the distribution of *yǐwéi* vs. *juéde* in naturalistic speech, we can use the same Human Simulation Paradigm to individually control and vary these variables to infer their impact on the acquisition of *yǐwéi*.

In addition, because our goal was to identify cues in children's learning inputs that could serve as "gems" for the acquisition of *yǐwéi*, we considered only child-directed speech sampled from the CHILDES datasets and were thus limited to 10 items per condition, resulting in analyses with limited power at the item level. One possible way to alleviate this issue is to artificially generate more stimuli. Future studies can explore this strategy or use additional adult-directed speech corpora to scale up the stimulus set.

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